Kingdom of Saudi Arabia
Ministry of Education
Umm Al-Qura University
031

المملكة العربية السعودية
وزارة التعليم
جامعة أم القرى
٠٣١

الاختبار النصفي لمقرر التعلم العميق المتقدم (AI4204)

Midterm Exam – Advance Deep Learning (AI4203)

مدة الاختبار	تاريخ الاختبار	الفصل الدراسي
ساعة	الأربعاء 1447/10/20 الموافق 2026/04/08	472

تعليمات الاختبار:

- عدد أسئلة الاختبار ١٢ سؤالا.
- درجة الاختبار ٣٠ درجة،
- تأكد من كتابة اسمك ورقمك الجامعي ورقم الشعبة في ورقة الإجابة .
- غير مسموح باستخدام أي مذكرات أو مراجع أثناء الاختبار (Closed book exam).
- في حال احتجت إلى مساعدة أو استفسار، ارفع يدك وانتظر المراقب للرد علي ك.
- احرص على متابعة وقت الاختبار لكي تتمكن من الإجابة على جميع الأسئلة
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Exam Instructions

- The exam consists of **12 questions.**
- The total exam score is **30 marks.**
- Make sure to clearly write your **name, university ID number, and section number** on the answer sheet.
- **No notes or reference materials are allowed** during the exam (**closed-book exam**).
- At the end of the exam, **submit both the question paper and the answer sheet** to the invigilator.
- If you need assistance or have a question, **raise your hand and wait for the invigilator** to assist you.
- Please **manage your time carefully** to ensure you are able to answer all required questions.

رقم الشعبة	اسم الطالب	الرقم الجامعي
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1. What are the three main parts that make up the Transformer architecture?

- Attention

- Embeddings

- Positional encoding

وين ال Encoder
ثم Decoder

2. What is the primary function of "Positional Encoding" in Transformers?

- Helps us to keep track of words order in a sentence

عبدالرحمن ال PE يتيح للنموذج فهم موقع كل كلمة في الجملة لأنه النموذج يعمل بالتوازي
ولا يفهم الترتيب تلقائية

3. In calculating self-attention, three variables taken from database terminology are used. What are they?

- Q → Queries

- K → Keys

- V → Values

4. In the self-attention mechanism, what does the Query (Q) represent?

A) The stored information retrieved as a result of the attention process

B) The current word that is "asking" which other words are most relevant to it

C) A label used to index and match words in the attention lookup table

D) The scaling factor applied before the SoftMax to stabilize gradients

5. The SoftMax function in the self-attention equation converts similarity scores into probabilities that sum to 1, producing attention weights for each word.

A) True

B) False

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6. What is the key difference between Self-Attention and Masked Self-Attention regarding access to other words?

A) Self-Attention can only look at previous words, while Masked Self-Attention can look at all words

B) Self-Attention allows every word to attend to all other words, while Masked Self-Attention restricts each word to attend only to previous words

C) Both allow full access to all words, but Masked Self-Attention applies dropout for regularization

D) Masked Self-Attention attends to future words only, while Self-Attention attends to past words only

7. Why do we transpose the Key matrix (K) when multiplying it with the Query

A) To normalize the values and prevent large numbers

B) To make the matrix dimensions compatible for multiplication

C) To reduce the size of the matrix for faster computation

D) To apply positional encoding to the input

8. What is the title of the seminal 2017 paper that first introduced transformer architecture, and what specific task was the model originally designed to perform?

- Attention is all you need

- Machine translation, specifically from English to german

9. In Encoder-Decoder Attention, where do Q, K, V come from?

- The decoder uses the outputs from the encoder to calculate K, V then the Q are then calculated using Masked Self-attention, in the decoder

10. Mention two practical uses for the "Context Aware Embeddings" produced by Encoder-Only Transformers?

- In classification tasks

- Clustering similar documents

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11. Explain what Word Embedding and Positional Encoding are, and how they are combined to form the input of a Transformer model.

PE ← - Positional Encoding: an index for each word we have that helps us track word order

- Word Embedding: a representation of text that allows the Neural network to process text (semantic vector representation) ادق لغوياً

- First text is transformed into an encoded tokens of Embedding and then the PE is added to it, where the index is added to the embedding vector forming the input to transformers

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12. Compare Encoder-Only and Decoder-Only Transformers. Explain the type of attention used in each, their primary goals, and why ChatGPT is considered a Generative Model.

Aspect	Encoder-Only	Decoder-Only
Attention	Self-Attention: looks at after and before words of interest	Masked Self-attention each word attend to itself and previous tokens only
Goal	Build a deep contextual understanding of words	Generate text step by step
Example tasks	Fully context-Aware embeddings used in classification tasks	Auto regressive models used for content generation

(!!) شو هذا يا وجيه

Why ChatGPT is generative: Because it is a decoder-only transformer that generates text step-by-step

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~ Good Luck ~

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Questions Overview (Marks Distribution)

Q#	Question Type	Marks
Q1	Short Answer	2
Q2	Short Answer	2
Q3	Short Answer	2
Q4	MCQ	2
Q5	True/False	2
Q6	MCQ	2
Q7	MCQ	2
Q8	Short Answer	3
Q9	Short Answer	2
Q10	Short Answer	2
Q11	Essay	4
Q12	Essay	5
Total		30